

G. ARAVIND

✉ g.aravind@hotmail.com ✉ aravind.gopal.1998@gmail.com in aravind-gk aravind-gk aravind-gk.github.io Q G-Aravind

EDUCATION

Indian Institute of Science, Bangalore

Oct 2020 – Present

M.Tech in Artificial Intelligence

8.1/10

Heritage Institute of Technology, Kolkata

Aug 2016 – Jul 2020

B.Tech in Computer Science and Engineering

8.8 / 10

RELEVANT COURSEWORK

Foundational Courses: Pattern Recognition and Neural Networks, Data Structures, Computer Vision, Data Analytics

Advanced Courses: Reinforcement Learning, Advanced Deep Learning, Deep Learning for Natural Language Processing

Math Courses: Random Processes, Computational Linear Algebra, Computational Methods of Optimization

Online Courses: Machine Learning with Graphs (Stanford)

WORK EXPERIENCE

Mastercard | Data Scientist Intern

Jun 2021 – Jul 2021

- Interned with AI Garage team at Mastercard, which focuses on designing AI-based solutions for use-cases in payments
- Majority of my time as an intern was devoted to building models for predicting high-value financial transactions, and the time and category of future transactions for a customer
- Modeled transactions as Marked Temporal Point Processes and used it to make predictions about future payments

ACADEMIC PROJECTS

Adversarial Attacks on Recommendation System Models | Thesis Project | Prof. Shirish Shevade

(ongoing)

- Using hyperparameter tuning techniques to tune the graph data in order to drop the recommender's performance
- Attacking the user-item graph using meta-learning techniques, to reduce performance of collaborative filtering systems

Multimodal Transformers for Image Captioning | Prof. Sriram Ganapathy

(Sep 2021)

- Image captioning formulated as a multimodal translation task, where input image features are extracted using various encoder models, and fed to deep networks for caption generation
- Used multimodal Transformer to capture both intra-modal and inter-modal interactions in a unified attention block

C-LSTMs vs Transformers for Multi-class Text Classification | Prof. Shirish Shevade

(Aug 2021)

- Implemented C-LSTM architecture (a combination of CNN and deep LSTMs) for classifying text documents
- Used self-attention mechanism at output, and dynamic meta-embeddings at input to improve C-LSTM performance
- Also used Transformers for document classification, and experimented with encoder blocks and positional embeddings
- Used Deep Averaging Networks (DANs) as baseline and tried different word/bi-gram embeddings

PERSONAL PROJECTS

- Modeling the spread of COVID-19 in Karnataka, India
- Movie recommendation system using graph neural networks and collaborative filtering
- Dynamic meta-embeddings for improved sentence representations
- Duckworth-Lewis method for predicting target runs in cricket matches, using non-linear regression models
- Gradient based optimization of hyperparameters for linear models
- Image compression using principal component analysis

TECHNICAL SKILLS

Languages: Python, C, MATLAB, SQL

Technologies / Frameworks: PyTorch, TensorFlow, Scikit-Learn

ACHIEVEMENTS / EXTRACURRICULAR

- All India Rank 6th in GATE Computer Science exam, out of ~1,00,000 candidates
- Teaching Assistant for E2 202: Random Processes, Fall 2021
- Certified in "CODECHEF Certified Data Structures and Algorithms Programme", Advanced Level (Jan 2018)